

LUYAO NIU (SHE/HER)

PhD student, University of Southern California
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EDUCATION

University of Southern California

PhD in Transportation Systems Engineering | Aug. 2026 - Present

Peking University

M.Sc. in Urban Planning (Smart City and Big Data Program) | Sept. 2023 - Jun. 2026 | GPA: 3.75/4

Beijing Jiaotong University

B.Sc. in Traffic Engineering; Minor in Finance | Sept. 2019 - Jun. 2023 | GPA: 3.97/4 | Top 0.5% | Beijing Outstanding Graduate

SELECTED RESEARCH EXPERIENCE

Joint Design of Shared Autonomous Vehicles and Public Transit Systems for Tourist Cities

- Formulated an integrated Shared Autonomous Vehicles (SAV)-transit design problem accommodating heterogeneous demand types (normal trips with carpooling and private tour trips requiring privacy), balancing system efficiency and accessibility in tourist city contexts.
- Applied continuum approximation and network queuing theory to derive flow equilibrium equations; formulated mixed integer program minimizing total system cost (agency and user costs) to determine optimal SAV fleet size and FRT network layout.
- Achieved 15-25% total cost reduction in low-demand scenarios compared to transit-only baselines; analyzed sensitivity of design outcomes to private tour trip proportions and demand levels; paper published in Transportation Research Part C .

Optimized Path Planning and Traffic Flow Assignment Model Balancing User Equilibrium and System Optimum

- Developed a traffic assignment framework balancing user equilibrium and system optimum to mitigate congestion caused by homogeneous navigation recommendations; conducted SP survey and used discrete model to elicit heterogeneous preferences and constructed personalized alternative route sets for systemoptimal assignment.
- Validated framework through simulation experiments on Beijing road network, demonstrating reductions in total travel time and congestion index; won First Prize in National University Traffic Technology Competition.
- Currently advancing research by applying inverse reinforcement learning on mobile signaling and trajectory data to better identify individual preferences and travel intentions; extending framework to connected and autonomous vehicle scenarios with centralized routing control.

Event-CausNet: Unlocking Causal Knowledge from Text with Large Language Models for Reliable Spatio-Temporal Forecasting

- Addressed limitations in spatio-temporal GNNs during non-recurring traffic events (accidents, work zones) by developing a causal inference framework that uses LLMs to extract and quantify causal knowledge from unstructured event reports, building a Causal Knowledge Base through Average Treatment Effects estimation.
- Engineered a dual-stream architecture integrating the Causal Knowledge Base with GNN-LSTM networks via novel Causal Attention mechanism to adjust traffic forecasts based on event-driven causal factors.
- Led full research execution from problem formulation to experimental validation; evaluated the framework on real-world datasets; submitted to KDD 2027.

Behavioral Changes from Electric Vehicles Adoption: A Quasi-experimental Study

- Built large-scale quasi-experimental study using mobile signaling data to estimate causal impacts of EV adoption on travel behavior, overcoming limitations of small-sample survey methods in existing literature.
- Developed novel identification strategy leveraging APP usage logs (charging and brand-specific apps) to identify EV adopters as treatment group and construct matched control groups for Difference-in-Differences causal inference analysis.
- Revealed car usage rebound effects (strongest during off-peak and non-commute periods), increased active travel linked to charging infrastructure access, and metro substitution effects;

AskNearby: An LLM-based Application for Neighborhood Information Retrieval and Personalized Cognitive-map Recommendations

- Identified the Local Life Information Accessibility (LLIA) gap in 15-minute city initiatives, where existing location-based systems neglect spatial, temporal, and cognitive factors critical for hyperlocal decisionmaking and community engagement.
- Developed AskNearby, an LLM-powered community platform integrating a three-layer Retrieval-Augmented Generation (RAG) pipeline (combining graph-based, semantic-vector, and geographic retrieval) with a cognitive-map model encoding users' neighborhood familiarity and preferences for context-aware recommendations.
- Achieved significant improvements over LLM-based and map-based baselines in retrieval accuracy and recommendation quality through real-world deployments; demonstrated robust spatiotemporal grounding and cognitive-aware ranking capabilities; paper accepted for presentation by ACM SIGSPATIAL GeoAI'25.

Multi-Agent Coordination and Collision Avoidance for Unmanned Warehouse Systems

- Proposed a hierarchical scheduling framework modeling multi-AGV task allocation as a Colored Traveling Salesman Problem; integrated Mixed-Integer Linear Programming for global order optimization with A* algorithm for precise pathfinding.
- Designed a dynamic zoning and load-balancing mechanism based on Monte Carlo Tree Search and UCT, optimizing pallet-to-workstation assignment to alleviate local congestion and minimize empty-load travel distance.
- Constructed a deadlock-free collision avoidance policy using Deep Reinforcement Learning with CNNbased state representation, ensuring robust navigation in dense environments; won Second Prize in China MathorCup Modeling Competition.

WORK EXPERIENCE

AI Agent Engineering Intern | May - Aug. 2026

Xiaohongshu · Data Analytics

- Domain knowledge infrastructure: Built a domain-specific data analytics knowledge base, structuring analytical knowledge to support knowledge-grounded agent workflows.
- Agent architecture and development: Developed the agent framework and implemented data analysis workflows, integrating knowledge resources and domain capabilities into the agent architecture.
- End-to-end validation: Validated the complete agent workflow, evaluating execution behavior and analytical outputs to guide systematic iteration.
- Domain Skill lifecycle: Designed and built domain-specific Skills, covering evaluation, monitoring, and ongoing operations to support maintainable, reusable agent capabilities.

Research Assistant | Jan. 2025 - Present

Data Science Department, Duke Kunshan University | Suzhou, China

- Led and independently executed research projects focusing on spatiotemporal data mining, model development, and traffic flow forecasting.
- Independently completed two studies: one published in IEEE GLOBECOM and another submitted to KDD 2027 (Event-CausNet).

EXTRA-CURRICULAR ACTIVITIES

President, Student Union | Jun. 2021 - Jun. 2023

Beijing Jiaotong University | Beijing, China

Volunteer, Beijing 2022 Winter Olympics | Jan. 2022 - Apr. 2022

National Indoor Stadium | Beijing, China

SELECTED AWARDS & HONORS

National Scholarship (Highest undergraduate honor in China) Dec. 2020

First-class Academic Scholarship (Top 1%) Dec. 2020

Baosteel Outstanding Student Award (Only 500 recipients nationwide annually) Dec. 2022

First Prize, China University Traffic & Transportation Technology Competition Jul. 2022

Second Prize, China MathorCup Optimizing Mathematical Modeling Competition Apr. 2022

Honorable Mention, Mathematical Contest in Modeling (MCM/ICM) Apr. 2022

Third Prize, China College Mathematics Competition Apr. 2020

Third Prize, Baidu ERNIE Cup AI Entrepreneurship Competition Sep. 2024

SKILLS

Languages: Mandarin Chinese (Native); English (Fluent, TOEFL iBT 101); Spanish (Conversational)

Programming & Data: Python (Pytorch, Scikit-learn, Geopandas, Networkx), C, MATLAB, R, SQL

Research Methods: Optimization, Machine Learning, Deep Learning, Reinforcement Learning, LLMs

PUBLICATIONS

Journal

Shi, Z., Luo, S., **Niu, L.**, Shao, C. (2024). Joint Design of Public Transit and Autonomous Vehicle Mobility Systems for Tourist Cities Considering Abnormal Trips. *Transportation Research Part C*. doi: 10.1016/j.trc.2025.105179.

Niu, L., Wang, Y., He, S., Zhang, W.(2025). A Cross-Lingual Framework driven by Large Language Models for Spatial Identity Sensing from Social Media. (Under review)

Liu, C., **Niu, L.**, Wan, L., Zhang, W. (2025) Travel Behavior Shifts Following Electric Vehicle Adoption: Evidence from a Big Data-driven Quasi-experiment. (Research manuscript)

Zhang, W., Wang, Y., Huang, N., **Niu, L.**, Wu, Y., Niu, C. (2025) Large-scale longitudinal mobility data reveal a causal view of social influence's amplification after risk. (Submitted to *Nature Human Behavior*)

Niu, L., Xing, X., Liu, Z., Yao, E. (2025) Traffic assignment Model balancing User Preference and system optimum. (In preparation)

Conferences

Niu, L., Wang, Z., Liu., J., Boukerche, A., Sun, P. (2025). MF-AttnBiLSTM: Traffic Flow Prediction via Hybrid Signal Decomposition and Dual-Stream Temporal Attention Learning. *IEEE GLOBECOM 2025*, Oral Presentation.

Niu, L., Deng, Z., Li, B., Huang, N., Liu, R., Zhang, W.(2025). AskNearby: An LLM-based Application for Neighborhood Information Retrieval and Personalized Cognitive-map Recommendations. *ACM SIGSPATIAL GeoAI'25*, Oral Presentation. doi: 10.48550/arXiv.2512.02502

Niu, L., Wang, Y., He, S., Zhang, W. (2025). Evaluating the Causal Impacts on Spatial Satisfaction Sentiment of Shenzhen-Hong Kong Regional Integration Policies: An Interrupted Time Series Study Leveraging Large Language Models. *EUGEO 2025 Proceedings*, Oral Presentation.

Niu, L., Gu, J., Liu, C., Zhang, W. (2025). Why Public Transport Policies Work or Not? A Metaanalysis Based on the Behaviour Change Wheel. *AESOP 2025 Proceedings*, Oral Presentation. (In preparation for submission to *Transport Reviews*)

Niu, L., Wang, Z., Guan, S., Sun, P. (2025) Event-CausNet: Unlocking Causal Knowledge from Text with LLMs for Reliable Spatio-Temporal Forecasting. (Submitted to *KDD 2027*) doi: 10.48550/arXiv.2511.12769

Niu, L., Huang, N.(2025) ST-ProC: A Graph-Prototypical Framework for Robust Semi-Supervised Travel Mode Identification. (Preprint) doi: 10.48550/arXiv.2511.13702

Zhu, H., Zhang, W., Huang, N., Li, B., **Niu, L.**,...,Liu, X. (2024). PlanGPT: Enhancing urban planning with tailored language model and efficient retrieval. (Preprint) doi: 10.48550/arXiv:2402.19273